

ECCE148: Signal Processing Applications

Assignment 8: Digital High-pass Butterworth Filters

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1 Introduction

The objective of this homework assignment is to construct a simple procedure for the design of *digital high-pass Butterworth filters*. For simplicity and consistency, we use the *bilinear transformation* method for frequency mapping between the analog and discrete domains.

The input to the procedure will be the standard design specifications of the digital high-pass filter. The design specifications are:

- | | |
|---------------------------------------|---------------------------------|
| a. <i>Pass-band Frequency:</i> | $\Omega_p = \pm \frac{5\pi}{6}$ |
| b. <i>Max. Pass-band Attenuation:</i> | $\alpha_{max} = 0.5 \text{ dB}$ |
| c. <i>Stop-band Frequency:</i> | $\Omega_s = \pm \frac{2\pi}{3}$ |
| d. <i>Min. Stop-band Attenuation:</i> | $\alpha_{min} = 20 \text{ dB}$ |

2 Design Procedure

We first identify the equivalent digital low-pass problem. The key insight is the spectral transformation $z = -Z$, which maps $H_{HP}(e^{j\Omega}) = H_{LP}(e^{j(\pi-\Omega)})$. Then, we pre-warp the digital low-pass edges to analog low-pass edges using the inverse bilinear formula:

$$\omega = \beta \tan\left(\frac{\Omega}{2}\right)$$

We compute the low-pass design order n and cutoff ω_c from the Butterworth design equations, and construct the analog low-pass design $H(s)$ from its Butterworth poles, scaled to ω_c . Then, we apply the bilinear transform:

$$s = \beta \frac{z - 1}{z + 1}$$

to get digital low-pass filter $\hat{H}(z)$. After applying the transformation:

$$z = -Z$$

we will produce the digital high-pass filter $H(z)$. Then, we verify the frequency response at Ω_p and Ω_s .

3 Analog Low-Pass Design Specifications

Using the transformation:

$$\Omega_{LP} = \pi - \Omega_{HP}$$

we obtain the digital low-pass design specifications:

$$\Omega_p = \frac{\pi}{6}$$

$$\Omega_s = \frac{\pi}{3}$$

We then prewarp the frequencies with $\beta = 2$ using the transformation:

$$\omega = \beta \tan\left(\frac{\Omega}{2}\right)$$

Thus, obtaining the analog low-pass filter design specifications:

- a. *Pass-band Frequency:* $\omega_p = \pm 0.536$ rad/s
- b. *Max. Pass-band Attenuation:* $\alpha_{max} = 0.5$ dB
- c. *Stop-band Frequency:* $\omega_s = \pm 1.155$ rad/s
- d. *Min. Stop-band Attenuation:* $\alpha_{min} = 20$ dB

4 Order and Cut-Off Frequency

From the Butterworth order equation:

$$n \geq \frac{\log_{10}\left(\frac{10^{\alpha_{min}/10}-1}{10^{\alpha_{max}/10}-1}\right)}{2 \log_{10}(\omega_s/\omega_p)}$$

we substitute in our analog low-pass filter design specifications and obtain:

$$n \geq 4.3631$$

We round up to get our final order:

$$n = 5$$

We determine the upper and lower bounds of the cut-off frequency:

$$\frac{\omega_p}{(10^{\alpha_{max}/10} - 1)^{\frac{1}{2n}}} \leq \omega_0 \leq \frac{\omega_s}{(10^{\alpha_{min}/10} - 1)^{\frac{1}{2n}}}$$

$$0.6614 \text{ rad/s} \leq \omega_0 \leq 0.7293 \text{ rad/s}$$

We choose the lower bound of the cut-off frequency, yielding:

$$\omega_0 = 0.6614 \text{ rad/s}$$

5 Analog Low-Pass Filter $H(s)$

We first identify the poles of the transfer function. Since $n = 5$ is odd, we have:

$$s = \omega_0 e^{j \frac{2k\pi}{10}}$$

for $k = 0, 1, 2, \dots, 9$. We select the 5 poles on the left-half plane from the set of 10 (poles that will yield $\theta = \frac{2k\pi}{10}$ between $\frac{\pi}{2}$ and $\frac{3\pi}{2}$). We then formulate the transfer function:

$$H(s) = \frac{\omega_0^n}{(s - s_1)(s - s_2)(s - s_3)(s - s_4)(s - s_5)}$$

$$H(s) = \frac{0.127}{(s - 0.661e^{j \frac{2\pi}{10} 3})(s - 0.661e^{j \frac{2\pi}{10} 4})((s + 0.661)((s - 0.661e^{j \frac{2\pi}{10} 6})(s - 0.661e^{j \frac{2\pi}{10} 7}))}$$

Thus, we obtain the transfer function of the analog low-pass Butterworth filter:

$$H(s) = \frac{0.127}{s^5 + 2.14s^4 + 2.29s^3 + 1.514s^2 + 0.619s + 0.127}$$

6 Low-Pass Digital Filter $\hat{H}(z)$

Using $\beta = 2$, we have the bilinear transform:

$$s = \beta \frac{z - 1}{z + 1} = 2 \frac{z - 1}{z + 1}$$

We then apply the bilinear transform on the analog low-pass filter to obtain the equivalent digital low-pass Butterworth filter:

$$\hat{H}(z) = \frac{0.001376(1 + z^{-1})^5}{1 - 2.9422z^{-1} + 3.7338z^{-2} + 2.4807z^{-3} + 0.8541z^{-4} - 0.1210z^{-5}}$$

7 High-Pass Digital Filter $H(z)$

We apply the transformation $Z = -z$ on $\hat{H}(z)$ to obtain the transfer function of the high-pass digital filter. We flip the sign of every other numerator and denominator coefficient:

$$H(z) = \frac{0.001376(1 - z^{-1})^5}{1 + 2.9422z^{-1} + 3.7338z^{-2} + 2.4807z^{-3} + 0.8541z^{-4} + 0.1210z^{-5}}$$

8 Frequency Response Plot

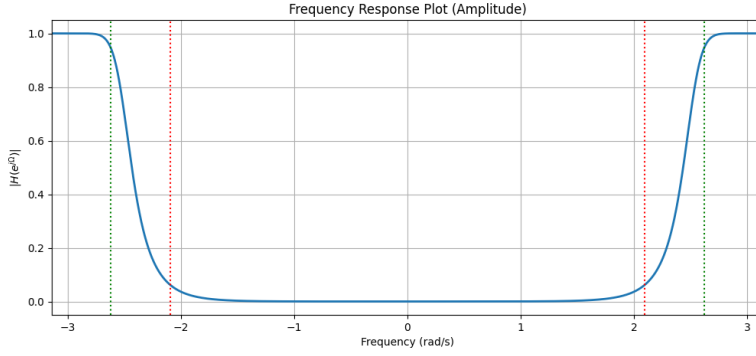


Figure 1: Frequency response plot (from $-\pi$ to $+\pi$, amplitude only).

In the pass-band region $\Omega_p = \frac{5\pi}{6}$, we have a near-unity gain. In the stop-band region $\Omega_s = \frac{2\pi}{3}$, we have strong attenuation.

9 Design Verification

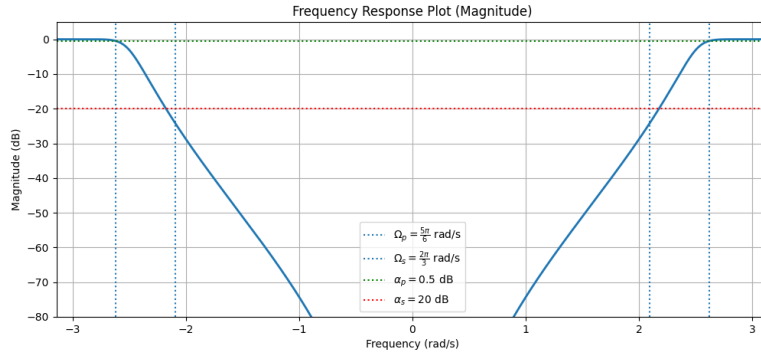


Figure 2: Frequency response plot (from $-\pi$ to $+\pi$, magnitude only).

We evaluate the frequency response at the pass-band frequency. At $\Omega_p = \frac{5\pi}{6}$, we find the magnitude to be 0.4978 dB, which is less than the maximum pass-band attenuation of 0.5 dB. At $\Omega_s = \frac{2\pi}{3}$, we find the magnitude to be 24.2169 dB, which is greater than the minimum stop-band attenuation of 20 dB. Thus, we verify that our design has passed the necessary specifications.